

Generative AI in IoT: Moving Towards Transformative Applications

Generative AI's contribution to the Internet of Things (IoT) cannot be underestimated. It has the potential to truly transform this domain by improving the security of IoT networks and making IoT devices more affordable, among other things.



Since the launch of ChatGPT 3.5 on November 30, 2022, no one seems to be able to stop talking or hearing about the buzzwords 'generative AI'. Generative AI is a type of artificial intelligence (AI) that can generate new content, such as text, images, audio, and video. It is trained on large data sets of existing content, and then uses this knowledge to create new and original content. Just like everything else, generative AI has the potential

to transform the Internet of Things (IoT) in several ways. For example, it can be used to create synthetic data for training machine learning models, generate personalised experiences for users, or improve anomaly detection, and much more.

Let's explore some of the specific ways in which generative AI is being used in IoT today, as well as review some of the potential future applications of this technology.

Creating synthetic data for machine learning

Leveraging generative AI to craft synthetic data for training machine learning models tackles a significant challenge in developing such models for IoT devices. While IoT devices can generate a lot of data, collecting and labelling it often proves expensive and time-consuming. Synthetic data originates artificially rather than from real world sources. It can be used to